

Chapter 29

Quick Reference

This chapter consolidates key equations, architecture specifications, API references, and failure mode diagnostics for rapid lookup during development and debugging.

29.1 Core RL & Alignment Equations

$$\text{PPO Clip: } L = \mathbb{E}[\min(r_t \hat{A}_t, \text{clip}(r_t, 1 \pm \epsilon) \hat{A}_t)], \quad r_t = \pi_\theta(a_t | s_t) / \pi_{\text{old}}(a_t | s_t) \quad (29.1)$$

$$\text{DPO: } L = -\mathbb{E}[\log \sigma(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)})] \quad (29.2)$$

$$\text{GRPO: } \hat{A}_i = (r_i - \mu_G) / \sigma_G, \quad \text{then PPO clip update (no critic)} \quad (29.3)$$

$$\text{KTO: } L = \lambda_w(1 - v(y_w)) + \lambda_l \cdot v(y_l), \quad v = \sigma(\beta \log(\pi_\theta / \pi_{\text{ref}}) - z) \quad (29.4)$$

$$\text{IPO: } L = \mathbb{E}[(\log(\pi_\theta(y_w) / \pi_{\text{ref}}(y_w)) - \log(\pi_\theta(y_l) / \pi_{\text{ref}}(y_l)) - 1 / (2\beta))^2] \quad (29.5)$$

$$\text{ORPO: } L = L_{\text{SFT}}(y_w) - \lambda \log \sigma(\log(\text{odds}(y_w) / \text{odds}(y_l))) \quad (29.6)$$

$$\text{GAE: } \hat{A}_t = \sum_{l=0}^{T-t} (\gamma \lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \quad (29.7)$$

$$\text{KL Penalty: } R_{\text{total}} = r_\phi(x, y) - \beta D_{\text{KL}}[\pi_\theta(y|x) \parallel \pi_{\text{ref}}(y|x)] \quad (29.8)$$

$$\text{RM (Bradley-Terry): } L = -\mathbb{E}[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))] \quad (29.9)$$

$$\text{Best-of-N: } y^* = \arg \max_{y_i \sim \pi_\theta(\cdot | x), i=1..N} r_\phi(x, y_i) \quad (29.10)$$

29.2 Transformer & Architecture Formulas

$$\text{Self-Attention: } \text{Attn}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k}) \cdot V \quad (29.11)$$

$$\text{Multi-Head: } \text{MHA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O, \quad \text{head}_i = \text{Attn}(XW_i^Q, XW_i^K, XW_i^V) \quad (29.12)$$

$$\text{RoPE: } f(x_m, m) = x_m e^{im\theta_j}, \quad \theta_j = 10000^{-2j/d} \quad (29.13)$$

$$\text{LoRA: } W' = W_0 + (\alpha/r) \cdot BA, \quad B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times k} \quad (29.14)$$

$$\text{KD (soft targets): } L_{\text{KD}} = (1-\alpha) L_{\text{CE}}(y, \hat{y}) + \alpha T^2 \cdot \text{KL}(p_T^{\text{teacher}} \parallel p_T^{\text{student}}) \quad (29.15)$$

$$\text{FFN (SwiGLU): } \text{FFN}(x) = (\text{Swish}(xW_1) \odot xW_3)W_2 \quad (29.16)$$

29.3 Decoding Methods

Method	Formula / Rule	Key Param
Greedy	$y_t = \arg \max_v P(v y_{<t})$	—
Beam search	Keep top- B partial sequences by joint probability	$B = 4-8$
Temperature	$P'(v) = \text{softmax}(\text{logit}_v/T)$	$T \in [0.1, 1.5]$
Top- k	Zero out all but top- k logits, renormalize	$k = 40-100$
Top- p (nucleus)	Keep smallest set V' s.t. $\sum_{v \in V'} P(v) \geq p$	$p = 0.9-0.95$
Min- p	Keep tokens with $P(v) \geq p_{\min} \cdot P(v_{\max})$	$p_{\min} = 0.05-0.1$
Repetition penalty	$\text{logit}_v \leftarrow \text{logit}_v/\theta$ if v appeared before	$\theta = 1.1-1.3$

29.4 Systems & Parallelism

Formula	Value (70B, BF16)	Description
Model memory	$2P$ bytes	140 GB (weights only)
Adam optimizer	$2P \times 4$ bytes (m + v)	280 GB
Full training footprint	$\sim 8P$ bytes	560 GB (weights + opt + grad)
FSDP memory/GPU	$8P/N_{\text{GPUs}}$	70 GB with 8 GPUs
Gen arithmetic intensity	$2P/2P = 1$ FLOP/byte	Heavily memory-bound
Token rate (gen)	$\text{HBM_BW} / (2P)$	~ 14 tok/s (A100, batch=1)
TP AllReduce / layer	$2 \times 2 \cdot \frac{T-1}{T} \cdot bsd$ bytes	~ 188 MB (70B, TP=8)
PP bubble fraction	$(P-1)/(P+M-1)$	P =stages, M =micro-batches
MFU	$\frac{\text{observed_toks}}{\text{peak_FLOPS}} \times 6P$	Target: $> 40\%$

29.5 GPU Hardware Specs

GPU	Memory	BW (HBM)	BF16 TFLOPS	NVLink	Notes
A100-80GB	80 GB HBM2e	2.0 TB/s	312	600 GB/s	Workhorse, widely available
H100-80GB	80 GB HBM3	3.35 TB/s	989	900 GB/s	Current gen, FP8 support
H200-141GB	141 GB HBM3e	4.8 TB/s	989	900 GB/s	Large context / fewer GPUs
B200	192 GB HBM3e	8.0 TB/s	2250	1800 GB/s	Next gen (2025)

29.6 Hyperparameter Ranges

Parameter	Typical Range	Default	Notes
β (DPO/KTO)	0.05–0.5	0.1	Higher = more conservative
ϵ (PPO clip)	0.1–0.3	0.2	Higher = more aggressive updates
γ (GAE discount)	0.99–1.0	1.0	Use 1.0 for episodic tasks
λ (GAE)	0.9–0.99	0.95	Lower = more biased, less variance
KL coeff (β_{KL})	0.01–0.2	0.05	Auto-adapt to target KL $\approx$ 5–8
LR (RLHF)	1e-7 – 5e-6	5e-7	Much lower than pre-training
LR (SFT)	1e-5 – 5e-5	2e-5	Standard fine-tuning range
LoRA rank r	8–128	16–64	Higher r = more capacity, more memory
LoRA alpha α	$r - 2r$	$2r$	Scaling factor; α/r is the effective scale
Temperature (gen)	0.6–1.0	0.7	Lower = less diverse candidates
Num generations K	4–64	4–16	For GRPO/Online DPO/Best-of-N
Grad clip norm	0.5–2.0	1.0	Prevents gradient explosion

29.7 TRL API Quick Reference

Trainer	Method	Key Config	Data Format
SFTTrainer	Supervised FT	packing, max_seq_length	prompt + completion
RewardTrainer	Reward model	center_rewards_coefficient	prompt + chosen + rejected
PPOTrainer	PPO	init_kl_coef, target_kl, cliprange	prompts (online gen)
DPOTrainer	DPO/IPO	beta, loss_type="sigmoid"/"ipo"	prompt + chosen + rejected
GRPOTrainer	GRPO	num_generations, beta, use_vllm	prompts + reward_fn
OnlineDPOTrainer	Online DPO	num_generations, reward_model_path	prompts (online gen)
KTOTrainer	KTO	desirable_weight, undesirable_weight	prompt + completion + label
ORPOTrainer	ORPO	beta	prompt + chosen + rejected
Best-of-N (manual)	Best-of-N	n_samples	prompts (inference)

29.8 RAG Pipeline Formulas

$$\text{Cosine similarity: } \text{sim}(q, d) = \frac{q \cdot d}{\|q\| \cdot \|d\|} \quad (29.17)$$

$$\text{Retrieval: } \mathcal{D}_k = \text{top-}k_{d \in \mathcal{C}} \text{sim}(\text{embed}(q), \text{embed}(d)) \quad (29.18)$$

$$\text{RAG generation: } P(y|q) = P_{\text{LLM}}(y \mid q, \mathcal{D}_k) \quad (29.19)$$

$$\text{Chunking overlap: } \text{stride} = \text{chunk_size} - \text{overlap} \quad (29.20)$$

$$\text{Reranker (cross-enc): } \text{score}(q, d) = \text{MLP}(\text{BERT}([q; d])) \quad (29.21)$$

29.9 Agentic Design Patterns

Pattern	Structure	Best For
ReAct	Think → Act → Observe → loop	General tool-use agents
Plan-and-Execute	Plan → Execute steps → Re-vise	Long-horizon, structured tasks
Supervisor	Router → specialist agents	Multi-domain, clear subtask boundaries
Swarm (handoffs)	Agent transfers control + context	Customer service, escalation flows
Hierarchical	Tree of delegating agents	Complex decomposition
Human-in-the-loop	Agent → Approval gate → Continue	High-stakes, irreversible actions

29.10 Agent Communication Protocols

Protocol	Scope	Transport	Key Concept
MCP	Tool integration	stdio / HTTP+SSE	Server exposes tools; client discovers & calls
A2A	Agent-to-agent	HTTP + JSON-RPC	Tasks with lifecycle (submitted→working→done)
OpenAI Function Calling	Tool use	API payload	JSON schema in <code>tools[]</code> array

29.11 Context Window Budget

$$C \geq \underbrace{S}_{\text{system}} + \underbrace{M}_{\text{memory/RAG}} + \underbrace{T}_{\text{tool defs}} + \underbrace{H}_{\text{history}} + \underbrace{R}_{\text{reserved output}} \quad (29.22)$$

Rule of thumb for 128K context:

- System prompt: 1–4K tokens (fixed)
- Tool definitions: 2–8K (scales with # tools)
- RAG context: 4–16K (top- k chunks)
- History: grows unbounded → summarize/truncate
- Reserved output: 2–8K

29.12 Common Failure Modes & Fixes

Symptom	Likely Cause	Fix
Reward up, quality down	Reward hacking	RM ensemble, length penalty, increase β
KL exploding (>15)	LR too high or mode collapse	Reduce LR, checkpoint rollback
Entropy collapse	Premature convergence	Increase entropy coeff, raise temperature
Training loss NaN	Gradient explosion	Reduce LR, increase grad clip, check data
No improvement after 5K steps	Bad prompt distribution	Goldilocks filter (20–80% pass rate)
Benchmark regression	Alignment tax	Reduce RL budget, use LoRA, mix SFT data
Length increasing monotonically	Length exploit in RM	Length penalty, retrain RM with length control
OOM during generation	KV cache overflow	Reduce batch, increase TP, PagedAttention
Agent loops forever	No max-iteration guard	Set <code>max_iterations</code> , add loop detection
Tool call parse failures	Inconsistent output format	Few-shot examples, constrained decoding
RAG returns irrelevant docs	Poor embedding / chunking	Reranker, hybrid search, smaller chunks
Multi-agent deadlock	Circular dependencies	DAG enforcement, timeout per agent

29.13 Method Selection Decision Tree

1. **Have paired preferences (chosen + rejected)?**
 - Noisy labels → **IPO**
 - Memory-constrained, no SFT done yet → **ORPO**
 - Clean data, limited compute → **DPO**
 - DPO plateaus, want exploration → **Online DPO**
2. **Have only binary feedback (thumbs up/down)?** → **KTO**
3. **Have verifiable rewards (math/code)?** → **GRPO**
4. **Need maximum quality, any cost?** → **PPO**
5. **Want training-free improvement?** → **Best-of-N**

29.14 Evaluation Metrics

Metric	Range	What It Measures
Perplexity	$[1, \infty)$	Model’s surprise; lower = better language modeling
Win Rate (vs. baseline)	$[0, 1]$	Fraction of outputs preferred by judge/human
BLEU	$[0, 1]$	n -gram overlap with reference (precision-focused)
ROUGE-L	$[0, 1]$	Longest common subsequence with reference
Pass@ k	$[0, 1]$	Probability ≥ 1 of k code samples passes tests
MMLU / GPQA	$[0, 1]$	Multi-choice accuracy on knowledge/reasoning benchmarks
HumanEval	$[0, 1]$	Functional correctness of generated code
Faithfulness (RAG)	$[0, 1]$	Fraction of claims supported by retrieved context
Context Relevancy	$[0, 1]$	Fraction of retrieved content relevant to query
Answer Relevancy	$[0, 1]$	Degree to which answer addresses the question

29.15 Reasoning & Test-Time Scaling

Method	Compute Cost	Mechanism
Chain-of-Thought (CoT)	$1.5\text{--}3 \times$ tokens	“Think step by step” in prompt
Self-Consistency	$N \times$ generation	Sample N CoT paths, majority vote on final answer
Tree-of-Thought (ToT)	$B \times D \times$ generation	BFS/DFS over reasoning tree; evaluate branches
Best-of- N	$N \times$ generation	Sample N , score with RM, pick highest
Beam search (on reasoning)	$B \times$ generation	Maintain top- B partial reasoning chains
Budget forcing	Variable	Allocate more tokens to harder problems dynamically
Verification (ORM/PRM)	$N \times$ gen + scoring	Generate N solutions, rank by outcome/process RM

29.16 Memory System Types

Type	Storage	Use Case
Working memory	Context window	Current conversation, immediate tool results
Episodic memory	Vector store	Past interactions, user preferences, session history
Semantic memory	Knowledge graph / embeddings	Facts, concepts, domain knowledge
Procedural memory	Skill library / code	How-to procedures, learned workflows

29.17 MCP Quick Reference

Primitive	Direction	Side Effects?	Purpose
Tools	Client → Server	Yes	Execute actions (create, modify, delete)
Resources	Client → Server	No (read-only)	Read data (files, DB records, configs)
Prompts	Client → Server	No	Reusable templates for common tasks
Sampling	Server → Client	No	Server requests LLM generation from client

Transport: `stdio` (local subprocess) or `HTTP+SSE` (remote, streamable).

Discovery: Client calls `tools/list`, `resources/list`, `prompts/list` at connection init.

Tool annotations: `readOnlyHint`, `destructiveHint`, `idempotentHint`, `openWorldHint`.

29.18 A2A Protocol Quick Reference

Concept	Description
Agent Card	JSON at <code>/.well-known/agent.json</code> — name, skills, supported content types
Task	Unit of work: <code>id</code> , <code>status</code> , <code>artifacts</code> . Lifecycle: <code>submitted</code> → <code>working</code> → <code>completed/failed</code>
Message	Communication unit within a task (role: <code>user/agent</code> , parts: <code>text/-file/data</code>)
Artifact	Output produced by the agent (structured data, files, generated content)
Push Notifications	Webhook-based updates for long-running tasks (via <code>tasks/pushNotification/set</code>)

Key endpoints: `tasks/send` (create/update), `tasks/get` (poll status), `tasks/sendSubscribe` (SSE stream).

29.19 Agent Framework Comparison

Framework	Orchestration	Multi-Agent	Best For
LangGraph	Explicit state graph	Conditional routing	Production: persistence, HITL, fine control
OpenAI Agents SDK	Declarative handoffs	Handoff-based	Simplicity: guardrails, tracing, fast start
AutoGen (AG2)	Conversation-driven	GroupChat	Prototyping: code execution, re-search
CrewAI	Role-based teams	Sequential/parallel	Low-code: quick demos, simple pipelines
Google ADK	Session + events	A2A native	Enterprise: artifact mgmt, multi-modal

29.20 Agentic RL Formulas

$$\text{Trajectory GRPO: } \hat{A}_i = (R(\tau_i) - \mu_G)/\sigma_G, \quad R(\tau_i) = \sum_t r_t^{(\tau_i)} \quad (29.23)$$

$$\text{Agent reward: } R = w_1 R_{\text{task}} + w_2 R_{\text{efficiency}} + w_3 R_{\text{safety}}, \quad R_{\text{eff}} = \max(0, 1 - \text{steps}/N_{\text{max}}) \quad (29.24)$$

$$\text{Masking: } \mathcal{L} = \sum_{t \in \text{agent tokens}} \min(r_t \hat{A}_t, \text{clip}(r_t) \hat{A}_t) \quad (\text{mask env outputs}) \quad (29.25)$$

$$\text{Pass@}k : 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}}, \quad n = \text{total samples}, c = \text{correct} \quad (29.26)$$

29.21 Agent Security Checklist

Threat	Layer	Mitigation
Prompt injection (direct)	Input	Input validation, instruction hierarchy, delimiters
Prompt injection (indirect)	Tool output	Treat tool output as untrusted; don’t follow instructions in retrieved docs
Tool misuse	Execution	Least-privilege permissions; <code>destructiveHint</code> gates; sandboxing
Data exfiltration	Output	Output filtering; restrict tool access to allowed domains
Excessive autonomy	Architecture	Max iterations; cost budgets; human approval gates
Confused deputy	Multi-agent	Verify task origin; capability-based access control

29.22 Agent Evaluation Metrics

Metric	Formula / Definition	Target
Task Success Rate (TSR)	Correct completions / total tasks	> 85% (production)
Steps to completion	Avg agent actions per successful task	Lower = more efficient
Cost per task	Total tokens × price/token	Budget-dependent
Latency (TTFC)	Time from request to first useful output	< 5s for interactive
Tool call accuracy	Correct tool selections / total calls	> 90%
Recovery rate	Successful retries / initial failures	> 60%
Human escalation rate	Tasks requiring human / total tasks	< 15%

29.23 Key Agentic Benchmarks

Benchmark	Domain	Metric	SOTA (2025)
SWE-bench Verified	Software engineering	% resolved issues	~70%
WebArena	Web browsing	Task success rate	~40%
OSWorld	Desktop computer use	Task success rate	~25%
GAIA	General AI assistant	Exact match accuracy	~75% (L1)
Tau-bench	Tool-use reliability	Pass rate (5 trials)	~65%
HumanEval / MBPP	Code generation	Pass@1	> 95%